



OLD DOMINION
UNIVERSITY



CCL Group Training

ECN Spray A CFD simulation

Date 09/21/2026



Overview of the ECN Spray A Target

The Engine Combustion Network (ECN) Spray A benchmark is a standardized high-pressure diesel spray experiment designed to provide high-fidelity target data for validating computational fluid dynamics (CFD) submodels under engine-relevant thermodynamic conditions.

Baseline Ambient & Injection Parameters

Injection Pressure

150 MPa

1500 bar fuel delivery

Ambient Temperature

900 K

High-temperature vessel

Charge Density

22.8 kg/m³

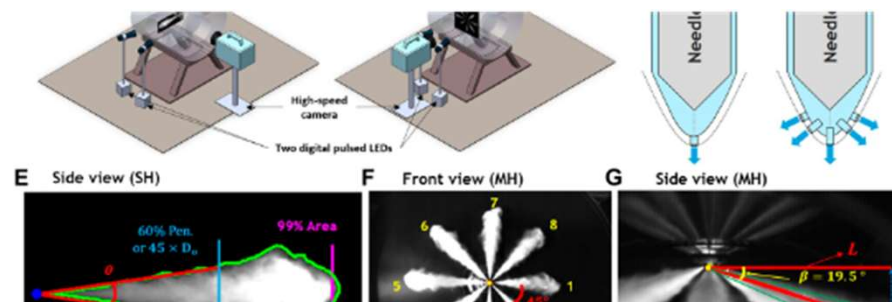
~6.0 MPa vessel pressure

Surrogate Fuel

n-Dodecane

Single-component C₁₂H₂₆

Conditions: 15% O₂ (Reacting) | 0% O₂ (Non-Reacting / Inert)



Primary Modeling & Validation Targets

-  **Nozzle Spec:** Single-hole axial orifice (90 μm) diameter, hydro-eroded).
-  **Spray Dynamics:** Liquid/vapor penetration length and dispersion angle.
-  **Combustion Metrics:** Auto-ignition delay time and flame lift-off length.



Physical Models Overview

Spray Dynamics

Lagrangian DPM

Breakup: Kelvin-Helmholtz & Rayleigh-Taylor (KH-RT).

Evaporation: Frossling correlation model.

Transport: Dynamic drag & NTC collision.

Turbulence

RANS / LES

RANS: Standard or RNG k-epsilon formulation.

Alternative: SST k-omega shear layer closure.

Jet Mixing: High-speed velocity gradient capture.

Combustion

SAGE Kinetics

Kinetics: Detailed SAGE chemistry solver.

Mechanism: Reduced n-dodecane chemical scheme.

Speedup: Adaptive multi-zone acceleration.

Autonomous Mesh

AMR & Embedding

Nozzle: Fixed embedding near injector tip.

AMR Criteria: Dynamic velocity gradient refinement.

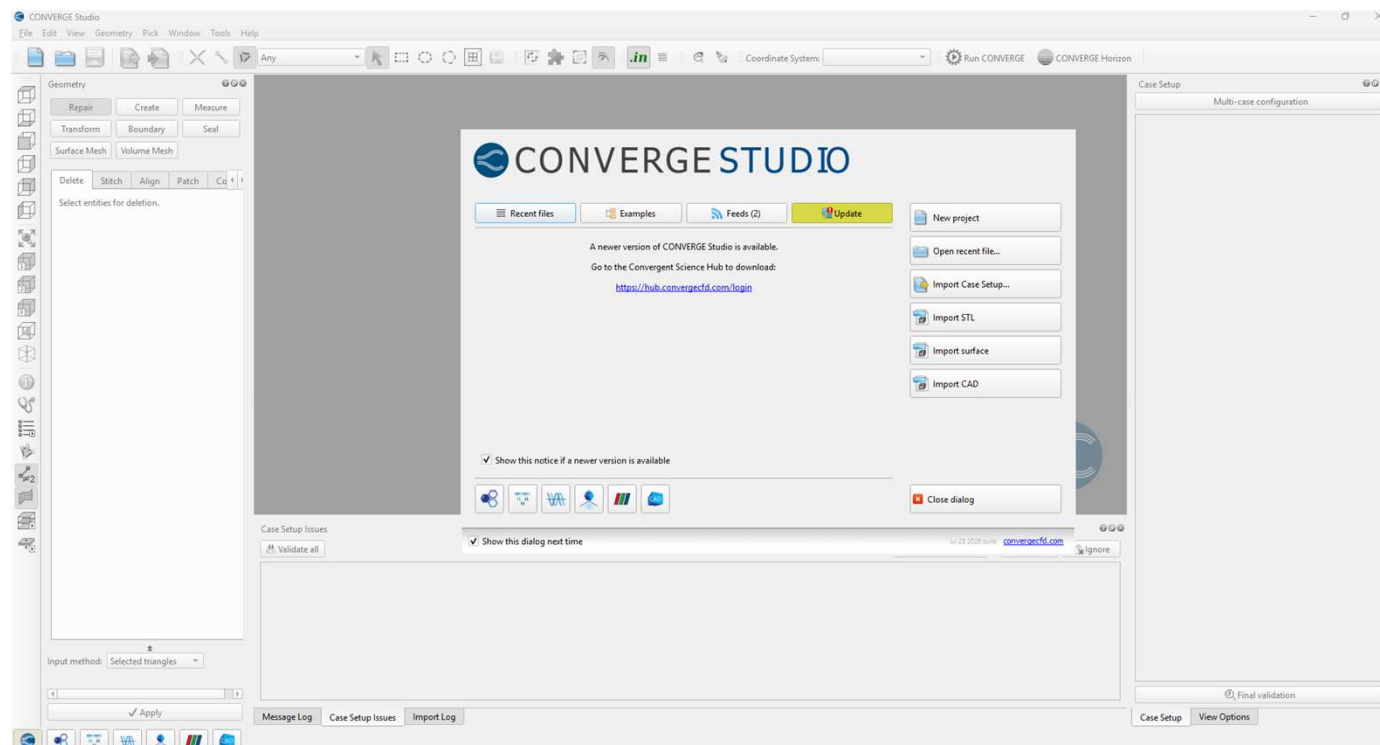
Flame Front: Temperature & species adaptive mesh.

Physics Integration: Fully Coupled Eulerian-Lagrangian Framework



Pre-Processing with CONVERGE Studio

CONVERGE Studio is the graphical user interface (GUI) and pre/post-processing suite for CONVERGE, a computational fluid dynamics (CFD) solver developed by Convergent Science.





SLURM Job Scripting & Execution

Slurm (Simple Linux Utility for Resource Management) is an open-source, highly scalable cluster management and job scheduling system used by high-performance computing (HPC) centers and supercomputers worldwide.

It acts as the "traffic controller" for an HPC cluster (such as ODU's Wahab cluster), coordinating how users share compute resources like CPUs, GPUs, and memory.

Core Functions:

- **Resource Allocation:** Allocates access to resources (compute nodes)
- **Job Scheduling:** Queues, prioritizes and manages pending execution requests.
- **Execution & Management:** Launches, monitors, and terminates tasks across allocated nodes.

```
#!/bin/bash
#SBATCH --job-name=my_job      # Name of the job
#SBATCH --partition=main      # Targeted cluster queue/partition
#SBATCH --ntasks=1            # Number of tasks
#SBATCH --cpus-per-task=4      # Number of CPU cores requested
#SBATCH --time=01:00:00        # Maximum runtime (1 hour)

# Executable commands run once nodes are allocated
mpiexec my_program
```



SLURM Job Scripting & Execution

Slurm Job Scripting for ECN Spray A simulation

```
#!/bin/bash
#SBATCH --job-name=ecnspray-a
#SBATCH --output=output_%j.log
#SBATCH --error=error_%j.log
#SBATCH --partition=main          # Use 'gpu' if you need GPUs, or 'timed-main' for short jobs (<2 hrs)
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=40
#SBATCH --time=01:00:00          # Walltime format: HH:MM:SS

export RLM_LICENSE=2765@vr-licmgr2.ts.odu.edu

export mpi=INTEL
export CVGHOME=/home/s3kim/apps/Software/CVG6/Convergent_Science/CONVERGE_CFD/6.0.2/

source $CVGHOME/environment/x64/scripts/CONVERGE/$mpi/6.0.2.sh

mpirun -n 1 converge-intel -l super >& logout.out
~
```



Post-Processing (ParaView Visualization)

ParaView is an open-source, multi-platform data analysis and visualization application used by scientists, engineers, and researchers to analyze extremely large datasets.

